Basic Linear Program

	Wheat	Corn	Beets
Yield (T/acre)	2.5	3	20
Planting cost ($/acre)	150	230	260
Selling price ($/T)	170	150	36 up to 6,000 T; 10 above
Purchase price ($/T)	238	210	–
Min. req. (T)	200	240	–

cision variables

x_1, x_2, x_3: acres to allocate to wheat / corn / beets

y_1, y_2: tons of wheat / corn to purchase

w_1, w_2, w_{3L}, w_{3H}: tons of wheat / corn / beets to sell (w_{3L}/w_{3H} quantity of beets sold in low / high quantity)

jective: max revenues – planting costs – purchasi

Max $170\, w_1 + 150\, w_2 + 36\, w_{3L} + 10\, w_{3H}$
$\quad - 150\, x_1 - 230\, x_2 - 260\, x_3 - 238\, y_1 - 210\, y_2$

nstraints

Land availability: $x_1 + x_2 + x_3 \le 500$

Wheat requirements: $2.5\, x_1 + y_1 - w_1 \ge 200$

Corn requirements: $3\, x_2 + y_2 - w_2 \ge 240$

Beets requirements: : $20\, x_3 - w_{3L} - w_{3H} \ge 0$

Beet quantity threshold: $w_{3L} \le 6{,}000$

Non-negativity: all variables ≥ 0

First-stage decision variables (do not depend on the scenario)

- x_1, x_2, x_3: acres to allocate to wheat / corn / beets

Second-stage decision variables (scenario-dependent)

- y_{1G} , y_{1A} , y_{1B}: tons of wheat to purchase in Good, Average, Bad scenario
- y_{2G} , y_{2A} , y_{2B}: tons of corn to purchase in Good, Average, Bad scenario
- w_{1G} , w_{1A} , w_{1B}: tons of wheat to sell in Good, Average, Bad scenario
- w_{2G} , w_{2A} , w_{2B}: tons of corn to sell in Good, Average, Bad scenario
- w_{3LG} , w_{3LA} , w_{3LB} (w_{3HG} , w_{3HA} , w_{3HB}) tons of beets to sell in low quantity (high quantity) in Good, Average, Bad scenario

jective

Max $1/3\,.\,(170\, w_{1G} + 150\, w_{2G} + 36\, w_{3LG} + 10\, w_{3HG} - 238\, y_{1G} - 210\, y_{2G})$
$\quad + 1/3\,.\,(170\, w_{1A} + 150\, w_{2A} + 36\, w_{3LA} + 10\, w_{3HA} - 238\, y_{1A} - 210\, y_{2A})$
$\quad + 1/3\,.\,(170\, w_{1B} + 150\, w_{2B} + 36\, w_{3LB} + 10\, w_{3HB} - 238\, y_{1B} - 210\, y_{2B})$
$\quad - 150\, x_1 - 230\, x_2 - 260\, x_3$ ← Land, first stage, fixed

nstraints across all scenarios

Land availability: $x_1 + x_2 + x_3 \le 500$

nstraints in Scenario "G"

Wheat requirements: $3\, x_1 + y_{1G} - w_{1G} \ge 200$

Corn requirements: $3.6\, x_2 + y_{2G} - w_{2G} \ge 240$

Beets requirements: $24\, x_3 - w_{3LG} - w_{3HG} \ge 0$

Sales price: $w_{3LG} \le 6{,}000$

nilar constraints in Scenarios "A" and "B"

n-negativity: all variables ≥ 0

Configure the → Observe → Sales

aircraft.

x_i $i=1,2,3$

uncertainty

d_1^s, d_2^s, d_3^s for $s=1,\dots,N$. (prob. p_s)

$y_{d_i}^s$ for $i=1,2,3$ $s=1,2,\dots,N$

Goal is to maximize expected revenue

$$\max_{x,y} \sum_{s=1}^{N} \underbrace{\left(\pi_1 y_1^s + \pi_2 y_2^s + \pi_3 y_3^s\right)}_{\text{revenue from scenario } s} p_s$$

s.t. $a_1 x_1 + a_2 x_2 + a_3 x_3 \le C$

$y_{d_i}^s \le x_i$, $y_{d_i}^s \le d_i^s$, $y_{d_i}^s \le x_2$ for $s=1,2,\dots,N$

Decision variables

- x_1: MWh of natural gas to procure per day
- x_2: MWh of solar energy to procure per day
- V: units of battery backup capacity to install

Model formulation

$$\begin{aligned} \min \quad & 3x_1 + 5x_2 + V \\ \text{s.t.} \quad & 200x_1 + 80x_2 \ge 160 \\ & 150x_1 + 500x_2 \ge 300 \\ & x_1 + x_2 \ge 2 \\ & x_1 + 2x_2 \le V \\ & x_1, x_2, V \ge 0 \end{aligned}$$

battery capacity V is the first-stage variable: The daily energy purchases are second-stage variables: they can be set each morning after observing the solar price and traffic forecast, so they are scenario-dependent.

- V : units of battery backup capacity to install (first-stage)
- x11: MWh of natural gas per day in Scenario 1
- x12: MWh of natural gas per day in Scenario 2
- x13: MWh of natural gas per day in Scenario 3
- x14: MWh of natural gas per day in Scenario 4
- x21: MWh of solar energy per day in Scenario 1
- x22: MWh of solar energy per day in Scenario 2
- x23: MWh of solar energy per day in Scenario 3

	A	B	C	D	E	F	G	H	I	J	K
		Probability									
	Scenario A	0.3									
	Scenario B	0.7									
		Stage 1		Stage 2 A		Stage 2 B					
		AquaSpa	HydroLux	AquaSpa	HydroLux	AquaSpa	HydroLux				
	Decision Variable	94	78	28	0	0	28				
								Sum Profit (Total by scenario)		Expected Profit	
	Profit Case A	350	300	350	300	0	0	66100		65540	
10	Profit Case B	300	350	0	0	300	350	65300			
11											
12	Scenario A							Sum (Used)	Constraint		
13	Pumps	1	1	1	1	0	0	200	200		
14	Tubing	9	6	9	6	0	0	1566	1566		
15	Labor	12	16	14	18	0	0	2768	2880		
16											
17	Scenario B										
18	Pumps	1	1	0	0	1	1	200	200		
19	Tubing	9	6	0	0	9	6	1482	1566		
20	Labor	12	16	0	0	14	18	2880	2880		
21											
22	Formulas	H9 = SUMPRODUCT(B7:G7,B9:G9)								Solver Simplex LP	
23		H10 =SUMPRODUCT(B7:G7,B10:G10)								Max: J9	
24		J9 =SUMPRODUCT(B2:B3,H9:H10)								by changing: B7:G7	
25		H13 =SUMPRODUCT($B$5:$G$5,B13:G13)				Copied to H13:H20				subject to: H13:H15 <= I13:I15	
26										H18:H20 <= I18:I20	
27										all variables >= 0	

Expected Profit is weighted average of all profit scenarios

	NPV	Cost y1	Cost y2	Cost y3	Cost y4	Cost y5
1	$141k	$75k	$25k	$20k	$15k	$10k
2	$187k	$90k	$35k	$0	$0	$30k
3	$121k	$60k	$15k	$15k	$15k	$15k
4	$83k	$30k	$20k	$10k	$5k	$5k
5	$265k	$100k	$25k	$20k	$20k	$20k
6	$127k	$50k	$20k	$10k	$30k	$40k

	Y1	Y2	Y3	Y4	Y5
Budget	$250k	$75k	$50k	$50k	$50k

Decision variables

- x_i=1 if project i is funded
- x_i=0 otherwise

Objective:

- max $141x_1 + 187x_2 + 121x_3 + 83x_4 + 265x_5 + 127x_6$

Constraints

- $75x_1 + 90x_2 + 60x_3 + 30x_4 + 100x_5 + 50x_6 \le 250$ (year 1)
- …
- $10x_1 + 30x_2 + 15x_3 + 5x_4 + 20x_5 + 40x_6 \le 50$ (year 5)
- x_i **binary** for all i

Binary Values (decisions) in a scheduling problem for example are parameters, not decision variables.
"Int" for integer variables, "bin" for binary variables

Binary: xi = {0,1} i = {1,2,3…7,8,n}
Budget: 250(cost)*x1 + 90x2 + 40x3 + … <= 250
Xi is binary!

Modeling Techniques:

Choosing at most 1 project: X1+…+xn <= 1
Select exactly k projects: x1+…+xn = k
Project 1 is a prereq for Proj 2: x1 >= x2
Project 1 AND 2 are prereq for Proj 3: X1 >= x3; x2 >= x3
Proj1 OR Proj2 are prereq for Proj3: x1+x2 >= x3
Select at le ast 1: x4+x5+x6>=1
Can only do Proj2 if Proj1 or Proj3 are chosen: X1 + x3 – x2 >= 0
Can only do Proj2 if both Proj1 and Proj3 are chosen: x1 + x3 – 2x3 >= (

Optimality Gap %: smaller %, better the solution but increased run time
Solving Limits: Larger limit gives better solution, longer run time.

14) X_i = # brate shows on machine i
Y_i = 1 if machine i is used

$$\text{Min} \quad \sum C_i X_i + \sum F_i Y_i \qquad \left(\begin{array}{l} C_i = \text{variable cost, machine } i \\ F_i = \text{fixed cost on machine } i \end{array}\right)$$

subject to

$X_1 + X_2 + X_3 + X_4 + X_5 + X_6 = 1800$

$X_i \le Cap_i$ (Cap_i = capacity of i)

$X_i \le 2000\, Y_i$

Y_i binary

* Suppose machines 1, 2 and 3 are different modes of the same physical machine, and at most 1 can be used:

$Y_1 + Y_2 + Y_3 \le 1$

Modeling Techniques:

Let x be decision variables: number of t-shirts to buy

Suppose the unit cost for the t-shirts are c, and there is a shipping cost f.

And there is **minimum order size of n**.

- Let y be the binary linking variable: $y = 1$ if you order any t-shirts
  - i.e. $y = 1$ if $x > 0$; $y = 0$ if $x = 0$
- Total cost
  - $total_cost = fy + cx$
- Constraints
  - $x \le My$ (M can be the maximum order size if there is one.)
  - $x \ge ny$
  - $x \ge 0$ and x is integer
  - y binary

Let y_1, y_2 be the binary linking variable:

- $y_1 = 1$ if you order any t-shirts
- $y_2 = 1$ if you order more than n_2 t-shirts

Let x_1 be the quantity at price c_1, and x_2 be the quantity at price c_2

Total cost

- $total_cost = fy_1 + c_1x_1 + c_2x_2$

Constraints

- $x_2 \le My_2$ (if $x_2 > 0$, $y_2 = 1$.)
- $x_1 \ge n_2y_2$ (if $y_2 = 1$, $x_1 \ge n_2$; if $y_2 = 0$, $x_1 \ge 0$.)
- $x_1 \le n_2y_1$ (if $x_1 > 0$, $y_1 \ge 0$.)
- $x_1 \ge n_1y_1$ (if $y_1 = 1$, $x_1 \ge n_1$.)
- $x_i \ge 0$ and x_i is integer for all i
- y_i binary for all i

– Presenters are indexed by $i = 1, \cdots, 14$ and time slots are indexed by $t = 1, \cdots, 6$ (e.g., $t = 1$ corresponds to the 9:00 am slot);
– We consider a parameter a_{it} equal to 1 if presenter i selected time slot t as a preferred time, and 0 otherwise (e.g., $a_{11} = 1$ and $a_{12} = 1$ as reported in Table 1).

Answer:

Decision variables

$$x_{it} = \begin{cases} 1 & \text{if presenter } i \text{ is scheduled in time slot } t \ (i = 1, \cdots, 14;\ t = 1, \cdots, 6) \\ 0 & \text{otherwise} \end{cases}$$

Model formulation

$$\begin{aligned} \max \quad & a_{11}x_{11} + a_{12}x_{12} + \cdots + a_{14,1}x_{14,1} \\ & \cdots + a_{1,6}x_{1,6} + a_{2,6}x_{2,6} + \cdots + a_{14,6}x_{14,6} && (1) \\ \text{s.t.} \quad & x_{11} + x_{12} + x_{13} + x_{14} + x_{15} + x_{16} = 1 && (2) \\ & \cdots \\ & x_{14,1} + x_{14,2} + x_{14,3} + x_{14,4} + x_{14,5} + x_{14,6} = 1 && (3) \\ & x_{11} + x_{21} + \cdots + x_{14,1} \le 3 && (4) \\ & x_{12} + x_{22} + \cdots + x_{14,2} \le 4 && (5) \\ & x_{13} + x_{23} + \cdots + x_{14,3} \le 2 && (6) \\ & x_{14} + x_{24} + \cdots + x_{14,4} \le 4 && (7) \\ & x_{15} + x_{25} + \cdots + x_{14,5} \le 3 && (8) \\ & x_{16} + x_{26} + \cdots + x_{14,6} \le 2 && (9) \\ & x_{it} \text{ binary, for each } i = 1, \cdots, 14;\ t = 1, \cdots, 6 && (10) \end{aligned}$$

The objective function (1) maximizes the total number of presenters scheduled during their preferred time slots. Constraints (2)–(3) ensure that each presenter is assigned to exactly one time slot. Constraints (4)–(9) ensure that the number of presentations scheduled in each slot does not exceed the available room capacity.

Is there a feasible solution to this problem? Why or why not? [5 pts]

Answer: Yes, there is a feasible solution. The total room capacity across all slots is 3 + 4 +

2 + 4 + 3 + 2 = 18, which exceeds the total number of presenters (14). Since every presenter

can be assigned to at least one slot without violating capacity, a feasible schedule exists.

	B	C	D	E	F	G	H
1							
2	Machine No.	1	2	3	4	5	6
3	Fixed Cost	1000	950	875	850	800	700
4	Variable Cost	21	23	25	24	20	26
5	Capacity	500	600	750	400	600	800
6							
7	Is it used?	1	1	0	1	1	0
8	Quantity Produced	500	600	0	100	600	0
9							
10	Cost	=C7*C3+C8*C4	=D7*D3+D8*D4	=E7*E3+E8*E4	=F7*F3+F8*F4	=G7*G3+G8*G4	=H7*H3+H8*H4
11							
12	Constraint	=2000*C7	=2000*D7	=2000*E7	=2000*F7	=2000*G7	=2000*H7
13							
14	Total Quantity	=SUM(C8:H8)					Total Cost
15							=SUM(C10:H10)

2,000 is Big-M value: upper bound to reasonably limit production capacity